

Generative AI for Physical Layer Communications: A Survey

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Abstract—The recent evolution of generative artificial intelligence (GAI) leads to the emergence of groundbreaking applications such as ChatGPT, which not only enhances the efficiency of digital content production, such as text, audio, video, or even network traffic data, but also enriches its diversity. Beyond digital content creation, GAI's capability in analyzing complex data distributions offers great potential for wireless communications, particularly amidst a rapid expansion of new physical layer communication technologies. For example, the diffusion model can learn input signal distributions and use them to improve the channel estimation accuracy, while the variational autoencoder can model channel distribution and infer latent variables for blind channel equalization. Therefore, this paper presents a comprehensive investigation of GAI's applications for communications at the physical layer, ranging from traditional issues, including signal classification, channel estimation, and equalization, to emerging topics, such as intelligent reflecting surfaces and joint source channel coding. We also compare GAI-enabled physical layer communications with those supported by traditional AI, highlighting GAI's inherent capabilities and unique contributions in these areas. Finally, the paper discusses

open issues and proposes several future research directions, laying a foundation for further exploration and advancement of GAI in physical layer communications.

Index Terms—Generative AI, physical layer communications, channel estimation and equalization, physical layer security, IRS, beamforming, joint source channel coding.

I. INTRODUCTION

THE RECENT surge in various large-scale datasets, combined with the ongoing progress in artificial intelligence (AI) technologies, has accelerated the development of generative AI (GAI) and led to the creation of GAI-based innovative applications such as DALL.E and ChatGPT [1]. The emergence of these killer applications has significantly enhanced the efficiency of digital content generation and enriched the variety of the produced content, signifying the arrival of the AI-generated content (AIGC) era [2]. Unlike traditional AI models, which focus mainly on analyzing, interpreting, and classifying data to solve specific problems, GAI excels in analyzing the distribution characteristics of complex data across different spaces and dimensions, uncovering data patterns [3]. On this basis, GAI can fully utilize the obtained features to generate outputs similar to its input data and present them to users in various forms. A representative example is stableDiffusion [4], which achieves state-of-the-art scores in class-conditional image synthesis and text-to-image conversion. Different from existing studies focusing on image classification or segmentation, stableDiffusion focuses on the generative, demonstrating greater flexibility and efficiency compared to traditional content creation techniques [5]. Through the fundamental working principles of GAI models and the representative examples, we can see that GAI possesses two core capabilities. The first is the ability to analyze and capture various features of complex data distributions. The second is the utilization of these captured features to generate new data that is similar to, but distinct from, the real data. Therefore, not only does GAI facilitate the generation of digital content, but its potent capability for data distribution analysis also supports research in various domains, including physical layer communications [6].

In wireless communications, a fundamental role of the physical layer communications involves converting digital data, generated by higher layers of the protocol stack, into

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a format suitable for transmitting over communication channels [7]. This process encompasses the steps of encoding the data into a bit sequence, modulating these bits onto a carrier wave, and then propagating the modulated signal through the channel. Correspondingly, at the receiver, this layer undertakes the inverse functions, i.e., demodulating the received signal, decoding the bit sequence, and forwarding the data to the higher layers for processing [8]. Beyond these core tasks, the physical layer is entrusted with several other key functions, such as channel access, channel equalization, and multiplexing [9]. Here, the channel access pertains to the process of determining which device is authorized to transmit data over the channel at any particular moment. Equalization involves compensating for the distortion and interference that can occur during transmission over a communication channel. Multiplexing, on the other hand, is the technique of amalgamating multiple data streams into a unified signal for channel transmission. Therefore, the physical layer is integral in shaping the overall reliability, effectiveness, and performance metrics of a wireless communication system [10].

Given its importance, researchers have conducted in-depth studies on the physical layer, including techniques such as beamforming [11], modulation and demodulation [12], signal detection [13], channel estimation [14], and channel state information (CSI) compression [15]. These techniques are directly linked to the analysis, compression, as well as the feature extraction of complex physical layer data. Conventional research relies on mathematically expressed models. However, in practical applications, the systems could include unknown effects that are almost impossible to be expressed analytically. Therefore, AI models have been introduced to support the physical layer functions of wireless communications. For instance, deep neural networks (DNNs) can learn the relationship between channel inputs and outputs to enhance the accuracy of channel estimation, thereby supporting the physical layer from various perspectives, such as signal detection, channel equalization, and synchronization [16]. In addition, deep learning (DL) models were also applied to support the physical layer communications. For example, recurrent neural networks (RNNs) can assist decoding [17], autoencoders can reduce peak-to-average power ratio [18], and Convolutional Neural Networks (CNNs) can compress the CSI in a massive multiple-input multiple-output (MIMO) system [19].

Although the traditional AI models are effective, their performance is limited. For example, DNNs can learn channel models, but they may struggle or even fail when dealing with *channels that are unknown during training*. Therefore, researchers introduce GAI, which can not only generate more channel samples to enhance the training data set but also assist in analyzing the distribution of existing data and extracting its key features, enhancing the system's capability to manage unknown channels [20], [21]. GAI can also improve physical layer security, beam forming, and other various physical layer techniques. However, applications of GAI have still not been well investigated, especially for emerging technologies such as intelligent reflecting surface (IRS) [22], cell-free, integrated sensing and communications (ISAC) [23], and extremely large-scale MIMO [24]. Therefore, further advancement of

GAI applications in the physical layer communications has been receiving a lot of attention recently.

Facing the emerging challenges in physical layer communications and considering the potential unique support offered by GAI, this paper provides a comprehensive survey of GAI's applications to address diverse problems in physical layer communications. We further discuss comparisons between techniques in the physical layer that are supported by GAI versus those relying on traditional AI models. We then discuss the lessons learned from existing studies, emphasizing the key capabilities of GAI employed in these instances. Lastly, the paper discusses open issues and future research directions. The key contributions of this paper are summarized as follows.

- We present the fundamentals of common GAI techniques, including generative adversarial networks (GANs), variational autoencoders (VAEs), normalizing flows, diffusion models, and transformers, as well as highlight their strengths, weaknesses, and differences. In addition, we discuss the special data generation properties of these GAI techniques that are particularly useful in solving various issues in the physical layer communications.
- We examine the problems of traditional AI-based solutions in the physical layer communications and illustrate how GAI can effectively address these problems. This reveals the unique support GAI can offer to the physical layer, beyond the capabilities of traditional AI, underscoring the importance of integrating GAI into physical layer techniques, particularly in dealing with various emerging technologies.
- We provide an in-depth analysis and summary of the GAI's applications in the physical layer communications, finding that these works primarily leverage three core capabilities of GAI. These include the ability to capture complex data distributions, the capability for cross-dimensional data transformation and processing, and the potential to repair and enhance data. This summary serves as vital guidance for further advancing the applications of GAI in the physical layer.
- We present significant open issues when applying GAI in the physical layer communications from several perspectives, such as privacy, security, and resource optimization, and provide some directions for future research.

The structure of this survey is outlined in Fig. 1. Section II offers a review of related works, while Section III delves into an in-depth analysis of existing studies. Section IV discusses open issues and future research directions, and Section V concludes the paper. Additionally, Table I lists the abbreviations widely used throughout this survey.

II. BACKGROUND AND RELATED WORK

This section discusses the background knowledge about GAI and some related surveys, and illustrates the differences between this survey and existing work.

A. Background of Generative AI

1) *Generative Adversarial Networks*: A GAN consists of two main elements, including (i) a generator that produces data

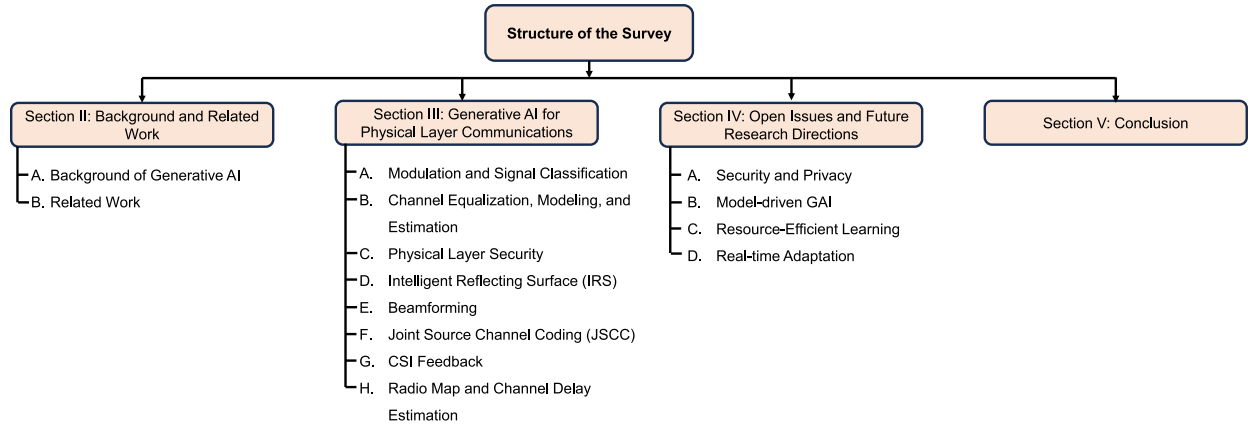


Fig. 1. The overall structure of this paper.

TABLE I
LIST OF ABBREVIATIONS

Abbreviation	Description	Abbreviation	Description
AI	Artificial Intelligence	DL	Deep Learning
TAI	Traditional Artificial Intelligence	GAI	Generative Artificial Intelligence
AIGC	AI-generated content	RNN	Recurrent Neural Network
DNN	Deep Neural Network	CNN	Convolutional Neural Network
CSI	Channel State Information	ML	Machine Learning
SNR	Signal-to-Noise Ratio	GAN	Generative Adversarial Network
BER	Bit Error Rate	PSK	Phase-Shift Keying
VAE	Variational Autoencoder	NF	Normalizing Flow
MIMO	Multiple-Input Multiple-Output	QPSK	Quadrature Phase-Shift Keying
mmWave	Millimeter Wave	UAV	Unmanned Aerial Vehicle
NMSE	Normalized Mean Square Error	PLS	Physical Layer Security
DRL	Deep Reinforcement Learning	RF	Radio Frequency
IRS	Intelligent Reflecting Surface	BS	Base Station
UE	User Equipment	FNN	Fully-connected Neural Network
JSCC	Joint Source Channel Coding	PSNR	Peak Signal-to-Noise Ratio
AWGN	Additive White Gaussian Noise	WGAN	Wasserstein GAN
SCMA	Sparse Code Multiple Access	MMSE	Minimum Mean Square Error

mimicking real data and (ii) a discriminator that differentiates between the real and generated data. The training process aims for a Nash equilibrium, where the discriminator cannot differentiate between the two [25]. Trained GANs are capable of reconstructing high-dimensional data from low-dimensional input with fewer generator function restrictions compared to other models, which makes them especially proficient in various issues in the physical layer communications such as channel estimation [26], CSI compression [27], and physical layer security [28]. Despite these advantages, GANs' training complexity lies in achieving the Nash equilibrium, which is more challenging than optimizing an objective function. This leads to the development of various GAN derivatives, such as StackGAN [29] and PAN [30], focusing either on architecture or objective function optimization [31]. These models have the great potential to effectively address various problems in the physical layer communications and wireless communications in general.

2) *Variational Autoencoders*: VAEs are neural networks designed for compressing and reconstructing data. They differ from traditional autoencoders by using probabilistic methods to model and generate data from a compressed latent space [32]. The VAE comprises an encoder that translates input data into a latent representation, and a decoder that

rebuilds the data from this latent space. These components are typically multi-layer neural networks. VAEs optimize their parameters by minimizing a loss function that assesses reconstruction accuracy and aligns the latent space distribution with a prior distribution. Key advantages of VAEs include their ease of implementation and training, effectiveness in learning compressed data representations, and a probabilistic nature that allows for uncertainty estimation and varied outputs [33]. As a result, VAEs are particularly effective in capturing the dynamics and uncertainty of wireless communications evidenced by a wide range of applications in channel estimation, channel modeling, and signal classification [34], [35]. In addition, the probabilistic nature of VAEs allows them to quickly learn robust data representations that are suitable for specific noisy channel conditions in joint source-channel coding [36]. However, they present challenges in training and parameter tuning, with the possibility of non-interpretable compressed representations.

3) *Normalizing Flows*: NFs are generative models that transform simple probability distributions into complex ones using reversible transformations. Unlike VAEs and GANs, they employ invertible neural networks for these transformations, which include a deterministic mapping function and an adjustable scaling and shifting function [37].

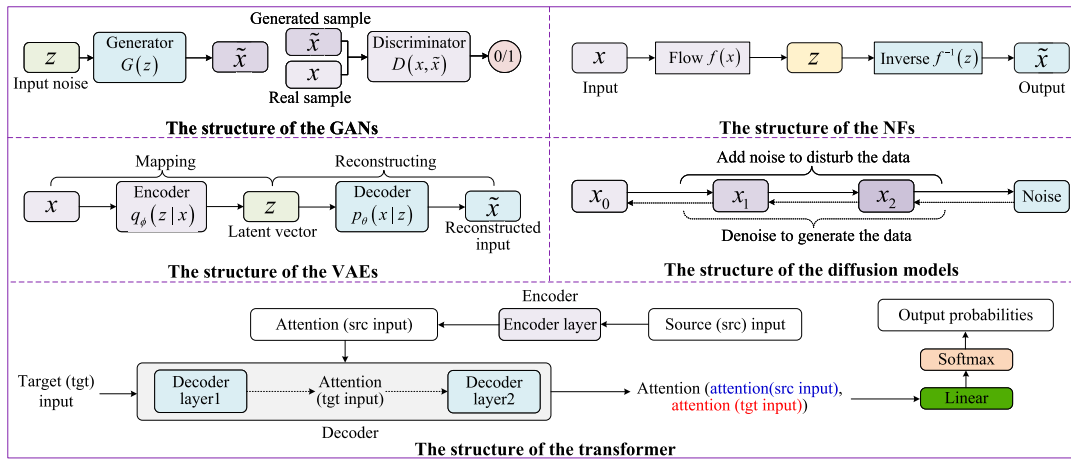


Fig. 2. The structure of common GAI models.

The representative examples are the Real NVP [38], which uses affine coupling layers and the Masked autoregressive flow [39], based on autoregressive models. The advantages of NFs lie in efficiently sampling complex distributions, managing high-dimensional data, and learning interpretable latent spaces. This is particularly useful for various tasks in the physical layer communications, especially under unknown noise distributions [40]. However, the challenges include high computational demands, lengthy training for complex distributions, and transformation function selection. Hence, recent studies have explored optimizing architectures and training efficiency through techniques such as adversarial training and regularization, demonstrating NFs' potential in diverse applications.

4) *Diffusion Models*: Unlike the aforementioned GAI models, diffusion models start with adding noise to training samples, which is known as the forward diffusion process, and then remove the noise to generate new samples in the inverse process [41]. They can be trained on incomplete data in a stable process. This special capability makes them highly suitable for equalizing and modeling wireless channels with limited training data and noisy conditions [42], [43]. However, diffusion models face challenges such as longer sampling times, complex training architectures, and limitations with certain data types [44]. To address these issues, researchers have developed optimization techniques, such as improving the training speed by reducing variance stochastic gradient descent, adaptive learning rate, and weight normalization.

5) *Transformers*: Another popular technique in GAI is transformers that can effectively learn sequential data, especially in natural language processing tasks. Unlike conventional sequence-to-sequence models that are based on CNNs and RNNs, transformers are designed by utilizing the self-attention mechanism [45]. In particular, the attention mechanism allows the model to attend all positions in the input sequence, learn their relations, and then compute a representation of the sequence. The transformer model consists of an encoder and a decoder. They are designed based on multiple self-attention mechanisms and feedforward neural networks. The encoder aims to process input sequence and capture

its complex dependencies while the decoder is responsible for generating the output sequence, corresponding to specific tasks. With the capability of learning long-range dependencies, transformers have been adopted in various tasks at the physical layers such as channel estimation, CSI feedback, and joint source channel coding [46], [47], [48]. Although having great potential for physical layer communications, the transformer architecture is still in its early stage of development with various challenges to be solved such as computational efficiency, efficient information injection, and model-driven integration [46].

In Fig. 2, we present the structures of the aforementioned GAI models. In addition, the principles, strengths, and weaknesses of these GAI models are summarized in Table II.

B. Relate Work

1) *Generative AI*: Given the GAI's growing popularity, numerous surveys have recently emerged. These surveys focus primarily on the fundamental architecture [49], [50], principles [51], implementation methods [52], as well as applications [3], [53], [54], [55], [56], [57] of GAI models. For instance, the authors in [50] provide a review on the GAI's history, basic components, and recent advances in AIGC across the unimodal and multimodal interactions. The authors in [51] present a survey on various deep GAI models, comparing these models, elucidating their underlying principles, interrelations, and reviewing current advancements and applications. For GAI's applications, the authors in [3] present a practical guide on using GAI for network optimization, demonstrating its effectiveness and contributing to network design. The authors in [55] examine GAI's role in the industrial Internet of Things (IIoT), focusing on the protection of trust-boundary and the prediction of network traffic, while highlighting challenges to accelerate its adoption. Differently, the authors in [57] discuss the applications of GAI in mobile telecommunications networks with a focus on problems such as spectrum sensing, channel analysis, network management, and network planning and deployment. Regarding the emerging Metaverse, the authors in [56] explore GAI's facilitative role in its

TABLE II
SUMMARY OF COMMON GAI MODELS

GAI Technique	Principle	Strengths	Weaknesses
GANs	Consist of a generator and a discriminator, training together through adversarial learning	High-quality and diverse generated samples	Hard to find Nash equilibrium during training
VAEs	An autoencoder that can learn the probabilistic distribution of a latent space and generate new data	Interpretable latent space and good for data reconstruction	Output of the model may be difficult to interpret
NFs	Transform simple probability distributions into complex ones using reversible transformations	Efficiently sampling complex distributions, managing high-dimensional data, and learning interpretable latent spaces	Training can be computationally expensive
Diffusion Models	Gradually add noise to training samples and then remove the noise to generate new samples in the inverse process	High-quality samples and learning stability	Long sampling times, time-consuming training, and limited diversity
Transformers	Attention-based architecture for sequence-to-sequence tasks	Learn long-range dependencies	High demands for memory and computation

TABLE III
SUMMARY OF THE RELATED WORKS

Ref.	Issue	Key focus of survey
[49]	Generative AI	An overview of some GAI models and architectures, training procedures, and limitations of three typical GAI models.
[50]		A summary of the history and fundamental components of GAI, along with recent progress in AIGC involving unimodal and multimodal interactions.
[51]		The principles, interrelations, current advancements, and applications of several GAI models.
[52]		The algorithms and implementation methods of several GAI models, as well as some guidance on selecting GAI models.
[53]		Technological development of various AIGC and application of GAI in education and creativity content.
[54]		An exploration of the advantages and disadvantages of using ChatGPT in educational contexts and some limitations of the ChatGPT.
[55]		The state of the art of GAI models and their use in IIoT, such as trust-boundary protection, anomaly detection, and so forth.
[56]		GAI's applications in Metaverse, such as avatars, non-player characters, and virtual world generation, automatic digital twin, and so forth.
[57]		An extensive overview of recent challenges and developments in applying GAI within mobile communications networks.
[3]		A tutorial of using generative diffusion model in network optimization.
[58]	AI Enabled Physical Layer Communications	A survey of the recent advancements in DL and its application in signal compression and detection.
[10]		An investigation about the DL-based physical layer, including using DL to redesign the modules in the traditional communication system and replace the communication system with autoencoder-based architecture.
[59]		Discuss some new applications of DL in the physical layer and present an autoencoder-based physical layer communication system.
[60]		An overview of recent advances of DL's applications in coding by focusing on sequential codes and Turbo codes.
[63]		Examine the necessity of real-time DL in the physical layer and provide a summary of the current developments and their limitations.
[61]		A detailed examination of different DL and deep reinforcement learning (DRL) methods suited for physical layer security applications.
[62]		The integration of machine learning with the selection of relay nodes, antennas, and authentication processes.

development, providing a research roadmap and addressing ethical implications.

2) *AI Enabled Physical Layer Communications*: AI models are crucial for advancing physical layer communications, spurring numerous research surveys. These studies primarily concentrate on the application of DL in various domains, including signal detection and compression [58], coding [10], [59], [60], security [61], [62], and communication delay [63]. For instance, the authors in [60] survey recent advances in DL-based coding, focusing on enhancing the specific coding method using DL techniques. The authors in [61] offer a review of DL-based security techniques for addressing issues such as attack detection and authentication in 5G and

beyond networks. Given the importance of communication delays, authors in [63] discuss the need for real-time DL in the physical layer, summarizing the current advancements and limitations in this area. The aforementioned surveys are summarized in Table III. The existing surveys about AI-enabled physical layer technologies and GAI, as discussed, provide two critical insights.

- From the perspective of GAI, the existing surveys primarily discuss the principles, architectures, implementation methods, and the strengths and weaknesses of different mainstream GAI models. Furthermore, researchers analyze the applications of GAI in general domains such as IIoT and mobile networks with a variety of

applications and provide future prospects and potential challenges from various aspects, such as ethical impacts and risks.

- Regarding AI-enabled physical layer communications, existing surveys review the physical layer technologies and conventional DL techniques. Besides, they present a detailed discussion of how these conventional DL techniques can support physical layer technologies, including signal compression and detection, coding theory, attack detection, physical layer authentication, and so forth.

Despite the comprehensiveness of these surveys, a gap remains in exploring GAI's applications in physical layer communications. Given the challenges posed by emerging technologies to the physical layer and the unique potential of GAI, this paper delves into how GAI can effectively address emerging issues in the physical layer communications. We further enhance this exploration by contrasting GAI-assisted physical layer technologies with those reliant on traditional AI models, thereby addressing current research gaps and providing insights into the ongoing evolution of GAI in physical layer communications.

III. GENERATIVE AI FOR PHYSICAL LAYER COMMUNICATIONS

In this section, we provide a comprehensive review of various applications of GAI for emerging issues in physical layer communications.

A. Modulation and Signal Classification

Signal classification and modulation recognition are always among the most important components in designing receivers of wireless communication systems [64]. However, traditional approaches usually require perfect or highly accurate knowledge of the underlying channel and CSI to obtain good detection performance [20]. Moreover, these approaches appear to be ineffective in future wireless communication systems due to the increased complexity of signals, spectrum efficiency requirements, and the dynamics of UEs' behaviors and characteristics. To overcome these challenges, DL is emerging as a prominent solution. Unfortunately, DL-based solutions require large datasets and long training time to obtain good detection performance [13], [65], especially when channel environments change fast due to user mobility. Moreover, a trained DL model only works well with some specific wireless environments that have similar characteristics to the trained environment. In new wireless environments with different conditions, e.g., channel models, surrounding interference, and noise distributions, this trained model will need to be retrained with a huge volume of new training data, which may not be feasible in practice. In addition, conventional DL-based solutions are less effective in modeling complex wireless channels that are time-varying, non-i.i.d distributed, or non-differentiable [20], [21]. To deal with these limitations, GAI, with its great capabilities to understand, capture, and generate the distribution of complex and high-dimensional data [66], [67], is a promising approach.

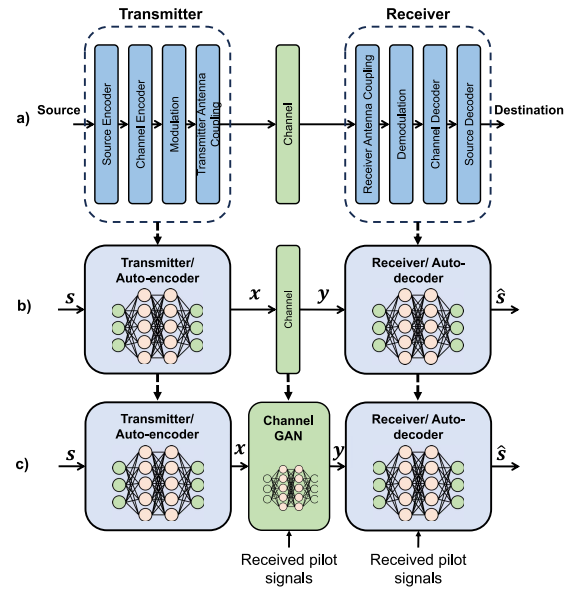


Fig. 3. Architectures of (a) traditional wireless systems, (b) end-to-end communication systems based on autoencoder, where the transceivers are represented by DNNs, and (c) end-to-end communication systems with channel GAN [21].

Specifically, the authors in [20] point out that traditional DL-based approaches do not perform well with non-Gaussian and time-varying channels, especially in the low signal-to-noise ratio (SNR) regions. For that, they propose a novel GAN to help the receiver intelligently adapt to the dynamics of wireless channels without retraining DNNs. In particular, the proposed GAN is used to efficiently learn the channel transition probability, i.e., the likelihood function. Then, the estimated channel transition probability is fed into the Viterbi algorithm [68] to derive the maximum-likelihood sequence detection. Moreover, the authors develop an online adjustment policy to fine-tune the proposed GAN network by leveraging the soft output of the model as well as pilot signals, making it more effective with time-varying wireless channels. The numerical results then demonstrate that the proposed GAN network can achieve a bit error rate (BER) of 10^{-2} at 8 dB SNR while the ViterbiNet approach [68] can only obtain this level of BER at 12 dB SNR. Moreover, the authors show that by using GAN they can obtain near-optimal BER performance under dynamic channel conditions.

Considering the same GAI method, the authors in [21] also develop a GAN network to model unknown channels in end-to-end wireless communication systems. As depicted in Fig. 3, the authors first consider an end-to-end communication system in which all the signal processing blocks at the transceivers are replaced by DNNs to jointly optimize the performance of the whole system. To do that, traditional DL-based approaches usually assume the availability of CSI and prior channel knowledge which are not always available in practice. To tackle this challenge, the authors design a novel conditional GAN network to represent the channel between the transmitter and the receiver to allow the gradient from the receiver to back-propagate to the transmitter. Moreover, the pilot signals received at the receiver are used as the conditional

information of the proposed GAN network, as illustrated in Fig. 3(c). In this way, the GAN network can generate more realistic coefficients for time-varying channels, and thus the end-to-end loss can be optimized to minimize the BER of the system. Interestingly, the authors demonstrate that the Kullback-Leibler divergence of the proposed GAN network can be significantly reduced when training the model over a long period, indicating that the generated data's distribution converges to the target distribution.

Besides signal classification, GAN can also be adopted for modulation recognition. For instance, the authors in [69] highlight that an application of DL for signal modulation recognition is often hindered by insufficient training data and overfitting. As such, the authors propose an auxiliary classifier GAN to enlarge the training dataset by generating new data while maintaining high-level features learned from the original training data. The authors then demonstrate that the proposed GAN solution can increase the classification accuracy by up to 6% compared to conventional DL-based solutions, e.g., AlexNet. Similarly, the authors in [70] propose a GAN network to restore up to 50% missing samples due to errors in dynamic spectrum sensing.

Differently, the authors in [71] propose a GAN-based modulation classification approach that is resilient to adversarial attacks. Specifically, the authors indicate that conventional DL-based automatic modulation recognition methods are vulnerable to adversarial attacks with well-designed perturbation injected into wireless channels. To tackle this practical issue, the authors propose a novel GAN network to generate plausible samples that are similar to the received frames. The generated frames are then compared with the perturbed received signals to detect the true class of the modulated signals. Simulation results show that the proposed GAN model can significantly increase the accuracy of DL-based modulation recognition methods under adversarial attacks. For example, the recognition accuracy for 8 phase-shift keying (PSK) scenarios can be increased from 9% to around 70% by using the proposed GAN model.

While most existing works in the literature adopt GAN, VAEs and NFs have been gaining attention recently [34], [40], [74], [76] due to their capabilities in dealing with signals in the time domain. For instance, the authors in [34] consider the signal classification problem in MIMO orthogonal frequency division multiplexing with index modulation systems. In particular, due to the high complexity of calculating the posterior probability, the authors estimate the variational posterior probability by training an encoder to map input data to a latent distribution and training a decoder to estimate the inputs, making it more effective than conventional DL approaches in approximating complex posterior distributions. Simulation results then reveal that the proposed approach can obtain near-optimal maximum-likelihood performance.

By using the NF technique, the authors in [40] propose a novel signal detection framework, which is fully probabilistic, to approximate unknown noise distributions. Specifically, the authors consider the signal detection problem in MIMO systems with unknown statistical knowledge of noise which is very challenging for traditional DL-based approaches. The

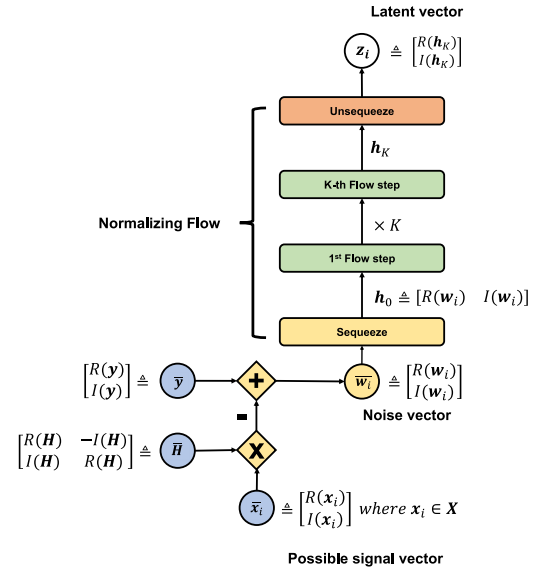


Fig. 4. Architecture of the NF-based detector with (i) a squeeze layer, (ii) K flow steps, and (iii) an unsqueeze layer. The noise vector for each signal vector is calculated and fed into the NF. After that, the NF's output is mapped into latent space z_i [40].

authors then utilize the NF technique to design a flexible detection framework that does not require any noise statistics as depicted in Fig. 4. The proposed NF is constructed by three major components, including an unsqueeze layer, K flow steps, and a squeeze layer. To obtain the maximum-likelihood estimation, the authors first calculate the noise vector $w_i = y - Hx_i$, corresponding to signal vector x_i with received signal y and channel matrix H . The proposed NF then maps w_i into the latent space which consists of latent variable z_i and the log-determinant. In this way, the corresponding likelihood $p(y|x_i)$ can be calculated, resulting in accurate maximum log-likelihood estimation. Extensive simulations demonstrate that the proposed framework outperforms existing DL-based methods in terms of BER under non-analytical noise settings. For example, in the quadrature phase-shift keying (QPSK) modulated 4×4 MIMO system, the proposed approach can reduce the detection error of the DetNet architecture [75] by 39.61% with SNR = 25 dB. However, the performance gap between the proposed method and the traditional maximum-likelihood approach is still noticeable. One potential solution is leveraging the auto-distribution technique to further improve the convergence of the proposed method in unknown noise conditions.

As summarized in Table IV, GAN is the most common GAI technique for modulation and signal classification tasks. This stems from its capability to effectively generate **high-quality synthetic samples** of wireless channels to significantly improve the training accuracy, especially when collecting real and labeled samples is difficult or even impossible. Besides GAN, VAE is also widely adopted due to its effectiveness in **estimating intractable posterior distributions** of wireless channels [74], [77]. Although possessing good detection performance, the dynamics and uncertainty of wireless environments have not been carefully investigated in existing

TABLE IV
SUMMARY OF GAI APPROACHES FOR MODULATION RECOGNITION AND SIGNAL CLASSIFICATION

Issue	Ref.	Drawbacks of TAI	Proposed GAI Approach
Modulation Recognition	[69]	Poor performance due to <i>insufficient data and overfitting</i>	Using the auxiliary classifier GANs to enlarge training datasets. GAN solution can increase the classification accuracy by up to 6% compared to conventional DL-based solutions.
	[71]	Vulnerable to <i>adversarial attacks</i> with well-designed perturbation	Propose a novel GAN network that consists of four generators to improve the model's accuracy and robustness against adversarial attacks
	[70], [72]	Require a <i>large amount of training data</i> , and the training accuracy can be greatly affected by the training data's quality	Propose a GAN-based method to generate missing wireless signal samples
Signal Classification	[73]	Lack of clean training dataset. Information loss during the feature extraction process.	Use GAN to generate a large training dataset without requiring manual annotation. Discriminative model can improve the signal classification process
	[21]	Poor performance due to the <i>dynamics of wireless channels</i>	Propose a conditional GAN to represent channel effects and bridge transmitter and receiver for joint optimizations
	[34]	Not effective in estimating posterior distributions	Use VAE to approximate intractable posterior distributions
	[74]	<i>Enormous data labels</i> are required	Use VAE to simplify the maximum-likelihood estimation which contains latent variables
	[20]	Require a <i>huge volume of training data</i> . Not perform well when the underlying channel models are completely unknown	Use GAN to directly approximate the transition probability of the underlying wireless channel. GAN network can achieve a BER of 10^{-2} at 8 dB SNR while the ViterbiNet approach [68] only obtains this BER at 12 dB SNR.
	[40]	Performance is not guaranteed when the <i>noise statistics are unknown</i>	Leverage an NF to effectively learn the distribution of unknown noise. Can reduce the detection error of DetNet architecture [75] by 39.61% with SNR=25 dB
	[76]	Require more bandwidth resources	Use VAE as a probabilistic model to recover transmitted symbols
	[77]	Sub-optimal when separate source and channel coding for short block lengths	Use VAE as a probabilistic model to recover transmitted short-packet symbols

studies. As such, designing real-time adaptive GAI models needs to be taken into account.

B. Channel Equalization, Modeling, and Estimation

In wireless communications, channel equalization, modeling, and estimation play essential roles in helping the receiver detect the received signals more efficiently. Over the past few years, DL has been widely adopted for channel equalization, modeling, and estimation both in academia and industry [16], [78]. Unfortunately, DL-based approaches require a huge volume of labeled data to learn sufficient characteristics of a specific channel, and thus limiting their application in dynamic wireless environments with high levels of randomness and variability. In addition, standard neural networks can work well for discriminative tasks but perform poorly when modeling the full complexity of channel distributions. Finally, conventional DL-based methods use a general loss function that makes their predictions less accurate, especially in low SNR regions [79].

For that, GAI has been adopted widely recently for equalizing, modeling, and estimating wireless channels, as summarized in Table V. Compared with conventional DL techniques, GAI possesses several advantages. Specifically, GAI can *generate synthetic training data that is similar to the data it was trained on* for ML models of channel estimation. In addition, it can generate data following specific constraints or conditions and leverage data from a source system to generate training data for a target system. All these special features make GAI an ideal tool for channel modeling. Moreover, GAI can be used as an equalizer to learn the mapping from

distorted signals to transmitted signals as well as to *model the posterior distribution* of transmitted signals and then estimate clean signals from distorted observations at the receiver.

1) *Channel Equalization*: In [80], the authors develop a hybrid GAN and autoencoder approach for channel equalization of underwater wireless communications with one-bit quantization. Specifically, it is highlighted that underwater wireless communications are extremely vulnerable to severe channel fading caused by the scattering and absorption of underwater environments. Moreover, the strong nonlinearities of one-bit quantization can greatly affect communication reliability. Given these challenges, using conventional DL-based approaches, e.g., autoencoder, may not yield good communication performance. For that, the authors propose to integrate GAN into their autoencoder architecture to significantly improve the channel equalization performance as illustrated in Fig. 5. Specifically, input signal $\mathbf{s}$ is first encoded by the encoder and then quantized by the adaptive one-bit analog-to-digital converter to reduce the energy consumption of the receiver. The generator of the proposed GAN architecture is used to approximate the distribution of encoded signal $\mathbf{e}_r$ given quantized signal $\mathbf{q}$ as its input. The discriminator then can distinguish the real encoded signal $\mathbf{e}_r$ and synthetic encoded signal $\mathbf{e}_f$ produced by the generator. Finally, the decoder will be used to recover the transmitted signal. In this way, the authors can construct a generalized channel equalization to equalize the one-bit quantization's distortion as well as the severe channel fading of underwater environments.

Due to its capabilities in analyzing signals in the time domain, VAEs have been widely adopted for channel equalization recently. For example, the authors in [81] and [82]

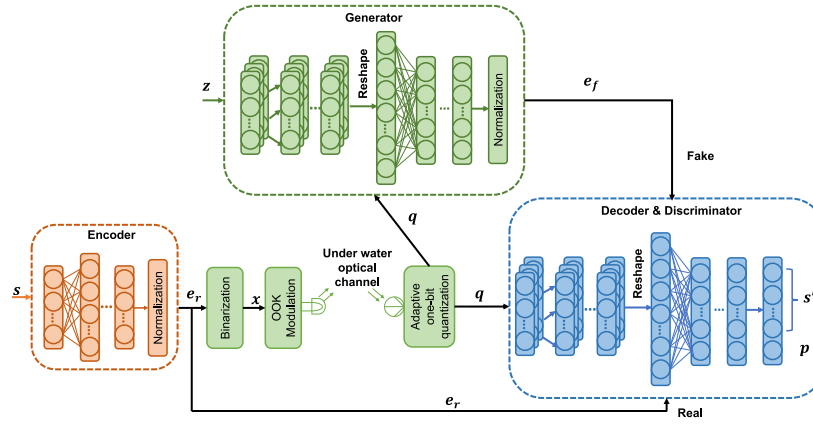


Fig. 5. Structure of the hybrid AE-GAN in which the encoder encodes input signal s , the generator learns the distribution of the real encoded signal e_r with the quantized signal q as its input, and the discriminator distinguishes the real encoded signal e_r and the fake encoded signal e_f generated by the generator [80].

propose to use VAEs for blind channel equalization which is challenging for conventional AI approaches. In particular, an encoder is used to represent the channel model and noise, and a decoder is used to approximate the posterior distribution of transmitted symbols from the received signals. In this way, the proposed VAE equalizer significantly outperforms baseline blind equalizers and obtains similar performance to that of a non-blind equalizer while not requiring prior knowledge of impulse responses and pilot signals. Differently, the authors in [43] propose to use a diffusion model to remove channel noise. In particular, the proposed channel denoising diffusion model is added as a new physical layer module right after the channel equalization to learn the input signals' distributions and then leverage them to further remove the channel noise. Experiments demonstrate that the proposed diffusion model can significantly reduce the mean square error and outperform existing approaches. For example, at SNR = 20 dB under Rayleigh fading, the proposed diffusion model can achieve a 1.06 dB gain compared to the joint source-channel coding system.

2) *Channel Modeling*: GAI also finds its applications in channel modeling [42], [83], [84]. In [83], the authors propose to use GAN to model millimeter wave (mmWave) channels. They highlight that accurately modeling mmWave channels is challenging due to several factors such as multiple high frequencies and highly directional beams. For that, the authors design a GAN approach to generate random profiles that include all information about the channel, including channel gains, delays, angle of arrival, and angle of departure of all links between the receiver and the transmitter. Simulation results then show that by using GAN, the authors can generate new channel data that have almost the same cumulative distribution function as real data. With this newly generated data, the authors then can effectively model mmWave channels by capturing the joint distribution of all links between the transmitter and the receiver with multiple frequencies.

In [84], the authors introduce a distributed GAN approach to model mmWave channels in unmanned aerial vehicle (UAV) networks. In particular, the authors state that existing approaches for channel modeling using conventional AI as

well as centralized GAN are limited by the lack of training channel samples and environmental measurements. For that, they propose to use UAVs to collect mmWave channel data during their aerial services. Each UAV employs GAN to train a local channel model. After that, the generated channel samples produced from the local channel model will be shared with other UAVs in the networks to improve their training process. Extensive simulations show that the proposed distributed GAN approach can significantly improve the modeling accuracy as well as increase the communication rate by 10% under real-time channel estimation compared to standalone training.

3) *Channel Estimation*: Besides channel equalization and channel modeling, GAI has been widely adopted in the literature for channel estimation. For example, the authors in [86] propose a GAN architecture for wideband channel estimation in mmWave and THz communications. The authors highlight that DL has been widely adopted for channel estimation in recent years. However, conventional DL-based approaches require long pilot sequences to achieve good estimation performance. Moreover, they provide poor channel estimation performance under high channel correlations and high propagation losses. For that, the authors propose to use GAN to estimate frequency selective channels at low SNR regions with short pilot sequences. Specifically, the proposed GAN approach can learn to generate realistic channel coefficients based on a real-world but unknown channel distribution during the offline training phase. After that, the trained GAN network is used as a prior model for online channel estimation by optimizing the input vector of the model based on the current signal received at the receiver. By doing this, the proposed GAN approach can obtain higher channel estimation accuracy with 70% fewer pilots compared to the traditional CNN networks (e.g., ResNet). Interestingly, the proposed solution can work well when changing the environment's factors such as the number of rays and clusters without retraining the GAN network.

Differently, the authors in [85] adopt GAN during the online training phase to further improve the channel estimation performance as illustrated in Fig. 6. Specifically, the receive sequence y_k and the training sequence p_k are fed into the

TABLE V
SUMMARY OF GAI APPROACHES FOR CHANNEL EQUALIZATION, MODELING, AND ESTIMATION

Issue	Ref.	Drawbacks of TAI	Proposed GAI Approach
Channel Equalization	[81]	Poor performance in <i>blind channel equalization</i>	Use VAE to efficiently learn from unknown input impulse sequences
	[82]	<i>High complexity</i> and not effective without using pilot symbols	Use VAE to design a blind channel equalization that can model the unknown nonlinearity
	[80]	Low performance under the scattering and <i>absorption effects of underwater communications</i>	GAN is used to equalize the one-bit quantization's distortion as well as the negative effects of underwater channels.
	[43]	Not effective in learning signal distributions	Use diffusion models to eliminate channel noise and achieve a 1.06 dB gain compared to the joint source-channel coding system
Channel Modeling	[88]	Only effective with simple channel models	Use GAN to learn the probability distribution functions of wireless channels, resulting in better channel response approximation
	[35]	Suffer from the <i>curse of dimensionality</i> and can only be evaluated with a simple AWGN channel model	Use VAE to learn the distribution of channel impulse responses and generate synthetic channel response samples with similar properties
	[83]	<i>Lack of training data</i>	Use GAN to generate random multi-cluster profiles that include all information of different frequencies
	[84]		Propose a distributed GAN architecture to allow UAVs to collaboratively approximate mmWave channel distributions and increase the communication rate by 10% under real-time channel estimation
	[42]	The collection of wireless channel data is <i>costly and time-consuming</i>	Propose a diffusion model based channel sampling approach to generate synthetic channel responses based on limited ground truth data
	[89]	Focus on estimating mmWave channel models for specific environments with limited applications	Use GAN for mmWave channel modeling by effectively extracting useful CSI features in the spatial-temporal domain
Channel Estimation	[90]	<i>High complexity</i> and training overhead needed to obtain channel knowledge	Use GAN to learn functions of channel covariance matrices and environment factors
	[26]	Cannot estimate channels in <i>high-speed moving scenarios</i>	Use GAN to learn and extract channel time-varying features and then restore channel information
	[86]	Need to know or model the channel distribution	Use GAN to generate synthetic channel samples that have a similar distribution with a true but unknown channel. Can obtain higher channel estimation accuracy with 70% fewer pilots compared to the traditional CNN networks
	[85]	Poor performance and require and require large datasets	Use GAN to learn from receive signals and exploit Wasserstein distance to improve estimation accuracy without transmitting long pilot sequences.
	[91]		Develop a conditional GAN approach to generate channel covariance matrices for training
	[87]	Low privacy due to large CSI dataset exchanging	Each client uses the estimated CSI obtained by the least square estimator as the input data of GAN to approximate the channel's distribution.
	[92]	Do not focus on the characteristic of mmWave frequencies or A2G wireless links	Use GAN to learn the distribution of mmWave channels from multiple distributed datasets
	[93]	<i>Not effective in high mobility scenarios</i>	Use Transformer to effectively track channel variation characteristics in highly dynamic environments
	[47]		Use Transformer to establish the parallel mapping between previous CSI and future channels
	[94]	Do not adequately account for the <i>dynamics and uncertainty</i> of channels in large MIMO systems	Use GAN to generate a more realistic channel image for more effective training under channel variations

proposed GAN architecture as its input data for training. The generator network then can “mimic” receive sequences $\hat{y}_k$ matching with the distribution of the true channel. After that, a newly proposed enhancement algorithm will perform channel estimation based on these new receive sequences. Simulation results indicate that the proposed GAN approach can help to improve the estimation accuracy of traditional training-based channel estimation approaches, especially at low SNRs.

The aforementioned GAN-based solutions and many others in the literature are designed in a centralized learning manner which may not be feasible in large-scale scenarios. To tackle this practical challenge, the authors in [87] propose a federated GAN solution for channel estimation in a distributed manner, as illustrated in Fig. 7. In particular, each client uses the estimated CSI obtained by the least square estimator as the input data of GAN to learn the distribution of channels. After

that, the generator parameters are transmitted to the server for aggregation. To improve federated learning performance, each client's discriminator will be dynamically adjusted by using regularizers based on the global generator's weights. Extensive simulations suggest that the proposed federated GAN approach is superior to conventional estimators as well as state-of-the-art DL-based channel estimation. For example, at SNR = 5 dB, GAN can achieve a normalized mean-squared error (NMSE) of 10^{-2} while the ChannelNet proposed in [95] can only obtain an NMSE of around 0.5. To further reduce the communication overhead, model compression and multiple tasks design can be considered to make the proposed federated GAN approach more effective.

Recently, the transformer model has attracted great attention in channel estimation due to its capability to learn long-range dependencies in data [47], [93]. For example, the authors

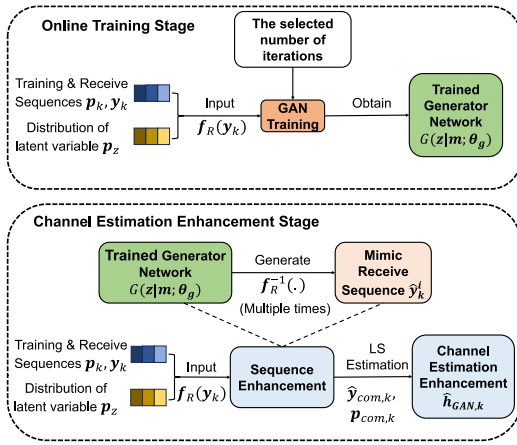


Fig. 6. Flowchart of GAN-based channel estimation. Training sequence p_k and receive sequence y_k are used as the input data. The generator then “mimics” receive sequences $\hat{y}_k$ matching with the distribution of the true channel [85].

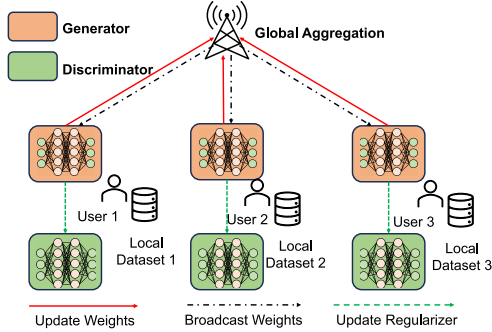


Fig. 7. Architecture of proposed federated learning, in which only the generators’ models are sent to the global server for aggregation. The discriminators then update their regularizer terms based on the global generator model’s weights [87].

in [47] develop a transformer-based parallel channel prediction approach to predict wireless channels in the next several frames simultaneously. In particular, the attention mechanism within the transformer model is used to establish the parallel mapping between previous CSI and future channels by performing simple matrix multiplications. In addition, the attention mechanism can add more weights to the previous CSI that are more useful in predicting future channels. As a result, the proposed transformer-based method can obtain better communication performance compared to conventional DL-based solutions. For example, with the user’s speed of 60 km/h, the proposed method can obtain a 10 dB NMSE performance gain compared with LSTM-based schemes.

In conclusion, GAI is particularly useful in channel equalization, modeling, and estimation thanks to its capabilities in **generating synthetic data** under constraints, **uncertainty estimation**, and **variational learning and sampling**. While GAN and VAE are the two most common GAI techniques for these problems, diffusion models and transformers can be also adopted as demonstrated in [42] and [47].

C. Physical Layer Security

Physical layer security (PLS) is another important research area in wireless communication systems. In general, PLS

refers to techniques that enhance the security of wireless communications at the physical layer by leveraging the inherent randomness of wireless communication channels. With recent advancements in DNNs, DL has been widely adopted to improve the security at the physical layer of wireless communication systems. However, conventional DL-based approaches are usually trained with specific environments, and thus cannot work well under attackers’ new strategies. In addition, it is difficult to collect sufficient labeled data from physical layer attacks due to their randomness and dynamics [28], [96], [97]. More importantly, conventional DL models are vulnerable to adversarial attacks [98], [99], in which minor perturbations in the input data can fool DNNs. Finally, they perform poorly with time-varying channels in low SNR regions and when prior information about attackers is not readily available [100], [101].

As discussed, GAI can be used for *anomaly detection* by generating data that is defined to be normal and then flagging input data that deviates significantly from these definitions. In addition, GAI has been demonstrated to be effective in *uncertainty estimation* as well as domain adaptation [103] which are critical capabilities to deal with physical layer security threats. For example, a GAN-based solution is proposed in [104] for abnormality detection at the physical layer in cognitive radio networks. In particular, the proposed GAN approach is used to generalize the state vectors extracted from spectrum representation data to learn the dynamic behavior of wideband signals. Based on these state vectors, abnormal signals can be distinguished from legitimate signals.

Similarly, the authors in [105] and [28] aim to prevent jamming attacks as well as interference from secondary users in cognitive radio networks. They first highlight that conventional DL-based anti-jamming approaches give poor performance when spectrum data is not sufficient. Unfortunately, collecting and labeling spectrum data in the presence of jamming attacks are time-consuming and costly. To address this practical issue, the authors propose to use GAN to generate synthetic spectrum data that can help a DRL algorithm to effectively learn and obtain the optimal dynamic spectrum anti-jamming access policy. Extensive simulations then demonstrate that the proposed GAN can help avoid complex jamming attacks and outperform conventional DRL-based approaches with incomplete spectrum information. The lack of training data problem of conventional physical layer security approaches is also discussed and addressed by using GAN in [97], [106], [100], and [107].

Differently, the work in [102] aims to authenticate radio frequency (RF) transmitters by using GAN. The authors first highlight that conventional ML techniques cannot be straightforwardly applied to RF systems due to the dynamics and uncertainty of RF signals. More importantly, these ML techniques may perform poorly in the presence of intelligent adversaries that can spoof transmitters and inject interference into the target channels, making it more challenging to capture the unique properties of the transmitters. For that, the authors propose a GAN-based approach to efficiently authenticate RF transmitters as GAN is well known for its capability in dealing with adversarial situations, as shown in Fig. 8. In

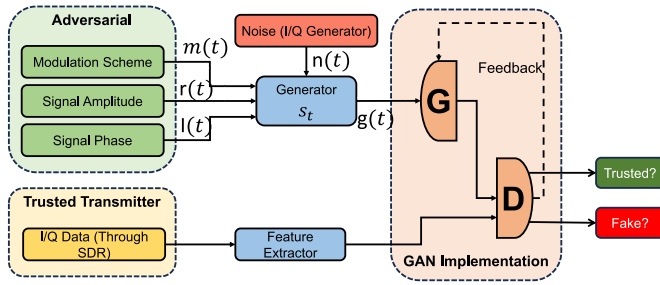


Fig. 8. GAN architecture for authenticating RF transmitters. Generator generate synthetic data $g(t)$ based on adversaries's signals. Discriminator takes input from Generator and "trusted" transmitters to distinguish real and fake signals [102].

particular, the GAN's generator will use RF signals generated from adversaries as its input to generate synthetic data $g(t)$. On the other hand, the discriminator learns from signals of both "trusted" transmitters and the generator to identify the differences between real and fake RF signals. In this way, the proposed solution can achieve a detection accuracy of 99% which is much higher than those of CNN and DNN approaches, i.e., 81.6% and 96.6%, respectively. Similarly, the authors in [108] also point out that the time-varying characteristics of wireless channels introduce more difficulties to conventional DL-based approaches in detecting abnormal RF signals. In contrast, GAN, with its capabilities of anomaly detection and uncertainty estimation, can deal with this issue effectively.

Besides GAN, VAEs can also be adopted for physical layer security. For instance, the authors in [101] propose a hierarchical VAE-based approach for physical layer authentication in complex scenarios such as industrial IoT systems. The authors state that ML has been widely adopted for physical layer authentication to analyze and extract complicated properties of wireless channels for authenticating wireless devices. Nevertheless, these methods usually require information about attackers available in advance to obtain good detection performance which is not the case in practice. As such, the authors develop a new hierarchical VAE architecture based on autoencoder and VAEs for efficient physical layer authentication with no prior channel information of attackers, as illustrated in Fig. 9. In particular, the VAE is used as a classifier, consisting of two hidden units Z_1 and Z_2 . Z_1 is constructed based on encoder ϕ_1 and decoder ψ_1 with a Simple Gaussian Prior for dimension reduction and channel impulse response reproduction. On the other hand, Z_2 is constructed based on encoder ϕ_2 and decoder ψ_2 with a revised double-peak Gaussian Prior for authentication. The conventional autoencoder is used to further reduce the dimension of input data. Finally, a new loss function is designed for the VAE module considering both the Simple Gaussian Prior and the double-peak Gaussian Prior distributions for further security and robustness enhancement. The authors then show that the proposed solution can improve the authentication performance by 17.18% compared to a conventional ML approach in [113].

Due to the ability to generate synthetic data that is similar to real data, GAI can also be used by adversaries to perform

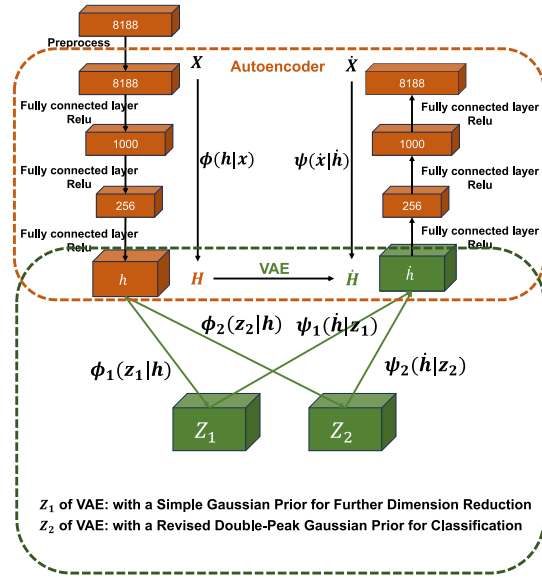


Fig. 9. GAN architecture for authenticating RF transmitters in which VAE is used as a high-level classifier with two hidden units: Z_1 is constructed with a Simple Gaussian Prior for dimension reduction and channel impulse response reproduction and Z_2 is constructed with a revised double-peak Gaussian Prior for authentication [102].

different types of physical layer attacks [98], [99], [116], [117]. For example, the authors in [116] use a GAN network to generate synthetic wireless signals that cannot be distinguished from legitimate signals by conventional approaches. Experiments then show that by using GAN the authors can improve the attack performance. Similarly, the authors in [99] recruit GAN to perform adversarial attacks. In particular, GAN is used to generate crafted imperceptible perturbations to cause wrong classifications of a DL-based modulation recognition approach. Through extensive simulations, the authors then indicate that the proposed GAN-based adversarial attack can reduce the accuracy of the DL-based modulation classifier more than jamming and other adversarial attacks. For instance, at 0 dB perturbation-to-noise ratio, the proposed techniques can reduce the detection performance by 37% at SNR = 10 dB, by 56% at SNR = 0 dB, and by 7% at SNR = -10 dB. In addition, the authors in [117] demonstrate that GAN can help a jammer to effectively jam a target wireless channel by generating more training data to help the jammer better learn the defense policy of the legitimate receiver. To deal with these GAN-based attacks, the authors in [98] propose to use another GAN network to augment the training dataset of the classifier with adversarial samples generated from adversaries' GAN networks. Simulation results then show that by augmenting the training data with GAN the authors can effectively improve the classification accuracy under GAN-based adversarial attacks.

As summarized in Table VI, GAN is mostly adopted to deal with threats in physical layer security. The reason is that the generator and discriminator networks can be trained through **adversarial training** to efficiently distinguish between real signals/data from trusted devices and fake signals/data from adversaries. Unfortunately, GAI can also be used by adversaries to perform attacks at the physical layer as GAI can

TABLE VI
SUMMARY OF GAI APPROACHES FOR PHYSICAL LAYER SECURITY

Issue	Ref.	Drawbacks of TAI	Proposed GAI Approach
Abnormality Detection	[109]	<i>Inaccurate representations</i> for the individual emitters	Use GAN to extract hidden information in original signals to improve identification performance
	[102]	Not effective when detecting rogue RF transmitters and classifying trusted ones	Use a generative model to generate fake signals that are trained together with real signals by a discriminative model to better identify trusted ones. Can achieve a detection accuracy of 99% which is much higher than those of CNN and DNN approaches, i.e., 81.6% and 96.6%, respectively
	[104]	Only effective with a specific type of low dimensional data	Use GAN to effectively learn from high dimensional data of spectrum representation samples
	[106]	<i>Lack of training data</i>	Use GAN to learn the data of trusted transmitters to extract RF fingerprint
	[107]		Use GAN to learn the distribution of collected signals
	[108]	<i>Time-varying characteristics</i> of wireless channels make the prediction unreliable.	Incorporating an encoder network into the original GAN to reconstruct the spectrogram
	[110]	Not effective in anomaly detection as GAI	Use GAN to identify unrecognized patterns on the model outputs and associated sequenced metadata
Authentication	[98]	TAI-based approach can be <i>cracked by using GAN</i>	Use GAN to augment the training dataset of the classifier with adversarial samples generated from another GAN network
	[111]	Not effective in learning distributions of received signals	Use GAN to learn the distribution of received channel data to authenticate a transmitting device
	[112]	Not effective in dealing with the <i>dynamics of wireless channels</i>	Use GAN and LSTM to learn and predict CSI elements' magnitude
	[101]	Need attackers' information for training	Use VAEs to extract valuable features of high-dimensional channel impulse responses for authentication. Improve the authentication performance by 17.18% compared to a conventional ML approach in [113]
Spoofing attack	[114]	Cannot generate high-quality synthetic spoofing signals	Use GAN to generate spoofing signals that are similar to legitimate signals
	[115]	Most detection methods cannot effectively detect spoofing jamming if spoofing signals are similar to authentic signals	Design a GAN network that is trained on a large dataset of authentic satellite signals to accurately learn their distribution
	[116]	Cannot effectively use to perform attacks	Use GAN to construct synthetic RF signals that are similar to legitimate signals
Jamming Attacks	[117]	Cannot generate high-quality synthetic samples	Use GAN to help jammers to generate training data to improve attack performance
	[118]	Cannot generate deceptive jamming templates under constraints	Use GAN to adaptively generate refined deceptive jamming templates based on various factors such as azimuth angles, angles, and target types. This can help to protect a specific area from observation and detection by adversarial radars
	[96]	<i>Lack of training data</i>	Use GAN to generate incomplete spectrum data in multiple jamming patterns
	[97]		Use GAN to generate more samples and use VAEs to learn the latent space of continuous signal samples.
	[105]	Poor performance when spectrum data is not sufficient	Use GAN to generate synthetic spectrum data that can help DRL to effectively learn and obtain the optimal dynamic spectrum anti-jamming access policy

effectively generate fake data that is similar to real data from legitimate activities. However, research on countermeasures against **GAI-based physical layer attacks** is still limited, and more efforts from both academia and industry are required.

D. Intelligent Reflecting Surface (IRS)

Recently, IRS has been emerging as a promising technology to significantly improve energy efficiency and spectrum utilization with low-cost and low-power hardware [119], [120], [121]. In particular, a typical IRS consists of a large number of reconfigurable metasurface elements that can be adjusted to reconfigure wireless channels and obtain high beamforming gain in a desired direction. However, accurate CSI information and underlying channel models must be obtained to leverage these advantages of IRS [79].

Unfortunately, it is challenging to acquire BS-IRS and IRS-UE channels separately without the help of RF chains. In addition, the cascaded channel of BS-IRS and IRS-UE links is very high-dimensional due to the high number of reflecting elements. To overcome these issues, various DL-based channel estimation and channel modeling approaches have been proposed in the literature. Nevertheless, these approaches cannot accurately estimate IRS channels since they use a general loss function that is not well designed for IRS, leading to poor estimation performance [79]. In addition, conventional DL-based approaches can only learn a limited number of channel parameters and one-dimensional channel impulse responses.

To tackle the above issues of conventional DL-based approaches, GAI has been adopted in various studies, as summarized in Table VII. For example, the authors in [119]

TABLE VII
SUMMARY OF GAI APPROACHES FOR INTELLIGENT REFLECTING SURFACE (IRS)

Issue	Ref.	Drawbacks of TAI	Proposed GAI Approach
Channel Modeling	[119]	Require in-depth domain knowledge and <i>lack of training data</i>	Use GAN to generate high-dimensional channel samples for training
Channel Estimation	[122]	Lack of observational dimensions and modeling capabilities	Combine GAN with a multiple residual dense network structure to effectively remove noise from estimated channel matrices. Achieve better NMSE performance and fast convergence compared with the residual network
	[123]	Not effective with <i>high channel dimensions</i>	Use GAN to learn the channel distribution with LS estimation as conditional input
	[79]	Use a general loss function that is difficult to make the estimated IRS channels more accurate	Use GAN to approximate cascaded channels by taking received signals as conditional information. NMSE of GAN is around ten times less than that of the ChannelNet architecture proposed in [124].
	[125]	The sparse cascaded channel assumption may not be valid in dense multipath propagation and non-line-of-sight settings.	Develop a graph transformer model to find useful correlations between different groups of IRS elements. Outperform a CNN-based approach with a 5 dB less SNR to achieve an approximate BER of 10^{-4}
IRS Deployment Design	[126]	Not effective in dealing with the <i>dynamics</i> of 6G networks	Use GAN to support DRL by learning the action-value that is near to target-action values, resulting in a more stable learning process. GAN architecture can help to improve the accuracy of DRL by 45%.

develop a model-driven framework based on GAN for channel modeling in IRS-aided wireless communication systems. To make GAN learn the channel distribution more effectively, the authors incorporate the structure of the cascaded BS-IRS and IRS-UE channels into the generator of the proposed GAN architecture. More specifically, the generative model now has three nodes: (i) BS-IRS node to learn BS-IRS channel distribution, (ii) IRS-UE node to learn IRS-UE channel distribution, and (iii) cascading node to combine the outputs. The discriminative model is then used to distinguish between the generated channel samples and the real BS-IRS-UE channel samples. Moreover, the authors adopt Wasserstein distance [127] to design a new loss function for the proposed GAN model for more stable training. In this way, the proposed solution can achieve much better performance than existing solutions using CNNs and fully-connected neural networks (FNNs), as demonstrated in the simulation results.

Differently, the authors in [79] use a conditional GAN architecture for channel estimation in IRS-aid wireless communications. In particular, the proposed GAN takes the received signals as its conditional information to generate channel responses with certain characteristics. Then, the discriminator and the generator compete with each other to obtain an adaptive loss function, making the generated channels similar to the original channels. With its capability to learn data distribution effectively, the proposed GAN architecture can achieve much better channel estimation performance compared to conventional DL-based methods as demonstrated in extensive simulations. For instance, at 5 dB SNR, the NMSE of GAN is around ten times less than that of the ChannelNet architecture proposed in [124]. Applications of GAN for channel estimation in IRS-aided wireless communications are also studied in [122] and [123] where GAN-based convolutional blind denoising and conditional GAN are adopted to obtain accurate CSI for IRS-aided systems, respectively.

GAN can also be used for the deployment design and phase shift optimization of IRS. For instance, the authors in [126] aim to jointly optimize the placement and reflecting beamforming matrix of an IRS-assisted 6G network. The

authors first develop a deep reinforcement learning (DRL) framework to interact with the system and gradually learn an optimal joint policy. However, due to the reward function's randomness, the proposed DRL framework cannot learn all the dynamics and uncertainty of the considered IRS system effectively. To overcome this issue, the authors propose to use GAN to identify the action-value that is close to target-action values, resulting in a more stable learning process. Specifically, the generator aims to generate actions (e.g., adjusting phase shift, coordinates, and beamform) for the DRL agent that are mapped to the original dataset. Then, these generated experiences and the original dataset are stored in a relay buffer. After that, the discriminator randomly takes a number of samples in the relay buffer as its input to learn how to distinguish the generated experiences from the generator and real samples from the original datasets. Simulation results then demonstrate that the proposed GAN architecture can help to improve the accuracy of DRL by 45%. To allow multiple IRSs to work collaboratively, the proposed approach can be extended by considering a multiagent GAN-based DRL framework.

Unlike the above studies, the authors in [125] propose a novel channel estimation method for IRS by leveraging the transformer model. Specifically, the authors first divide the IRS surface into various groups and use the least square approach to estimate the channels for some groups. Then, a graph transformer model is developed to estimate the channel for the remaining groups. The attention mechanism inside the transformer model aims to find useful correlations between different groups based on the channel information obtained by the least square detector. As such, the authors can estimate the channels for the remaining groups with lower training overhead compared to existing methods. Experimental results demonstrate that the proposed transformer-based method can outperform a CNN-based approach with a 5 dB less SNR to achieve an approximate BER of 10^{-4} .

As summarized in Table VII, GAI can be used to deal with various issues in IRS such as channel modeling, channel estimation, and IRS deployment. Existing studies in the

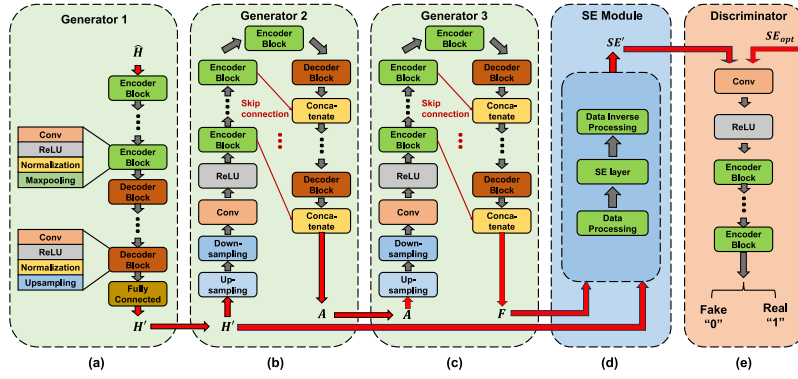


Fig. 10. Multi-GAN architecture for beamforming with (a) Generator G1, (b) Generator G2, (c) Generator G3, (d) the spectrum efficiency module, and (e) the discriminator [130].

TABLE VIII
SUMMARY OF GAI APPROACHES FOR BEAMFORMING

Issue	Ref.	Drawbacks of TAI	Proposed GAI Approach
Beamforming	[131]	Lack of training data	Use GAN to generate additional data for beam prediction
	[128], [132]	Not effective in learning data distribution	Use VAE to approximate the probabilistic model of beam dynamics
	[129]	High feedback overhead and high complexity	Use GAN to reconstruct a low-dimensional channel fed back from the receiver to perform hybrid beamforming at the transmitter. Reduce 2,048 complex channel elements to just 15 real values while maintaining good communication performance
	[130]	Suffer from the rank-deficient problem	Use a GAN architecture with three generators to recover (i) rank-deficient channels, obtain (ii) analog beamforming and (iii) digital beamforming matrices. Improve the beamforming performance by 47.49% compared to a conventional CNN-based method

literature mainly focus on GAN-based solutions. Nevertheless, other GAI techniques such as VAEs, normalizing flows, and transformers also have the great potential to further improve the performance of IRS as they can **efficiently learn the underlying distribution** and **capture the complex dependencies** of the IRS cascaded channel. This opens new research directions of novel GAI-based solutions for IRS.

E. Beamforming

In wireless communications, beamforming is a key technology to improve signal quality and transmission coverage. However, it is challenging to obtain optimal beamforming policies due to the high computational complexity and excessive feedback overhead, especially in systems with large antenna arrays such as mmWave and massive MIMO communication systems [128], [129]. DL can be used to tackle this problem but it requires a large amount of training data and cannot efficiently deal with the dynamics and uncertainty of wireless communications. Several researchers have been adopting GAI as an alternative approach and achieving promising results, as summarized in Table VIII.

For example, the authors in [129] propose to use GAN to reconstruct low-dimensional channel feedback from the receiver to perform hybrid beamforming at the transmitter, resulting in low communication overhead. Specifically, the generator of the proposed GAN architecture is first pretrained offline with channel samples generated by a geometric channel model to learn the channel structure and correlations. In

the online phase, the receiver tries to compress the channel matrix to a low-dimensional vector and feeds it back to the proposed GAN architecture at the transmitter to recover the channel matrix that will be used for beamforming design. The proposed GAN solution can help to reduce 2,048 complex channel elements to just 15 real values while maintaining good communication performance as demonstrated in simulations. One possible extension of this work is to extend the considered model/method to extremely-large massive MIMO or holographic MIMO.

Differently, the authors in [130] consider the beamforming design in massive MIMO systems with large antenna arrays where only rank-deficient CSI can be obtained. This rank-deficient problem has not been fully solved in the literature using conventional techniques. For that, the authors propose a multi-GAN architecture for hybrid beamforming design under rank-deficient channels. Specifically, the authors employ three generators in the proposed GAN architecture in which generator G1 is used to recover rank-deficient channels, generator G2 is used for analog beamforming, and generator G3 is used for hybrid beamforming, as illustrated in Fig. 10. Generator G2 takes the estimated rank-deficient channel $\mathbf{H}'$ as its input to generate analog beamforming $\mathbf{A}$ which is the input of generator G3. Generator G3 then estimates hybrid beamforming $\mathbf{F}$. After that, $\mathbf{H}'$ and $\mathbf{F}$ are fed into the spectrum efficiency module to calculate the average spectrum efficiency $\mathbf{SE}'$. The generator then learns from $\mathbf{SE}'$ and the real spectrum efficiency to improve the training processes of generators G1 and G3. Extensive simulations then demonstrate

TABLE IX
SUMMARY OF GAI APPROACHES FOR JOINT SOURCE CHANNEL CODING (JSCC)

Issue	Ref.	Drawbacks of TAI	Proposed GAI Approach
JSCC	[133]	Not effective in dealing with the <i>complexity and discontinuity</i> of source data distributions	VAE's encoder converts the source data into a low-dimensional latent space while the VAE's decoder tries to recover it to original data for JSCC. Increase the average PSNR by nearly 3 dB compared to conventional CNN-based approaches
	[134]	Not effective when <i>source dimension increases</i>	Use VAE to learn the source distribution by considering the noise channel as a sample of latent variables. Achieve up to 28.91% bandwidth saving
	[135]	Have significant losses of perceptual quality for the edge cases	Use two GAN-based networks to recover the distorted reconstructions of a DL-based JSCC and to produce the latent and noise inputs for the StyleGAN-2, respectively
	[36]	<i>Not stable</i> for multivariate Gaussian source over Gaussian multiple access channels	Propose a VAE-based JSCC system with added distribution restrictions on the loss function to avoid falling into the local minimum in specific regions
	[137]	Suffer from the <i>cliff effect</i>	Use diffusion models as a generative refinement component to enhance the reconstruction's perceptual quality
	[138]	<i>High complexity</i>	Use a GAN compression method based on intermediate feature distillation

that the proposed multi-GAN architecture can improve the beamforming performance by 47.49% compared to a conventional CNN-based method.

In conclusion, GAI is an effective approach for beamforming in wireless communications due to its effectiveness in **learning data distributions** and **generating high-quality samples**. GAN is mainly adopted in existing studies due to its effectiveness and simple architecture. Recently, VAE has emerged as an alternative approach for beamforming as it can efficiently approximate the probabilistic model of beam dynamics. Other GAI techniques such as normalizing flows and transformers should be considered in future studies for beamforming.

F. Joint Source Channel Coding (JSCC)

Coding plays a crucial role in wireless communications to mitigate the negative effects of channel noise, interference, and fading. Traditionally, the transmitter performs source coding for compression and channel coding for error correction, separately, making it difficult to optimize the spectrum usage. By combining the functions of source coding and channel coding into a single process, JSCC can leverage the statistical characteristics of the source and the channel to design a more efficient coding method. However, the complexity and discontinuity of source data distributions introduce challenges to the design of JSCC. For that, the authors in [133] propose a novel JSCC approach based on VAEs over additive noise analog channels. Specifically, the proposed VAE's encoder is used to convert source data into a low-dimensional latent space while the VAE's decoder recovers it to original data for JSCC. More importantly, the authors study that when the channel dimension is smaller than the source dimension, the encoding of two neighboring source samples needs to be near each other for good encoding performance. Therefore, multiple encoders are employed, and one of them will be selected for sample encoding on a specific side of the discontinuity. Experiments then demonstrate that using the proposed VAE-based JSCC method can help to increase the average peak SNR (PSNR) by nearly 3 dB compared to conventional CNN-based approaches.

Recently, JSCC has been emerging as an effective technology for semantic communications. However, in [134], the authors highlight that when the source dimension increases,

e.g., large-scale images, the performance of DL-based JSCC methods degrades significantly. Moreover, when the channel bandwidth ratio increases, these methods provide poor coding gain as they cannot learn the source distribution to determine patch-wise variable-length transmissions. To tackle these issues, the authors design a JSCC architecture based on VAEs in which the noise channel is viewed as a sample of latent variables. In this way, the proposed architecture can effectively learn the source distribution to provide a more effective coding mechanism. Experiments then show that the proposed solution can achieve up to 28.91% bandwidth saving or a PSNR gain of 2.64 dB on the CIFAR10 dataset while the conventional deep JSCC increases the bandwidth cost by up to 54.31%.

Similarly, the authors in [135] also consider JSCC for semantic image transmissions. The authors study that DL-based JSCC possesses significant perceptual quality losses in edge scenarios. Therefore, they propose two novel JSCC schemes based on GAN, namely InverseJSCC and GenerativeJSCC. InverseJSCC aims to recover the distorted reconstructions of a DL-based JSCC model via solving an inverse optimization problem using a pre-trained style-based GAN architecture. In contrast, in GenerativeJSCC, GAN is used as the decoder to produce latent and noise inputs for the StyleGAN-2 [136] generator. By jointly training the encoder and GAN decoder, GenerativeJSCC can outperform DL-based JSCC methods in terms of perceptual quality and distortion, as demonstrated by extensive simulations. GAI has also been adopted in other studies as summarized in Table IX.

As summarized in Table IX, VAE is mainly adopted in existing studies for JSCC. This is because VAE is particularly powerful in converting high-dimensional data into **low-dimensional latent space**. Moreover, the transformer, as a sequence-to-sequence model, can be a potential approach for JSCC to effectively learn the complex and **long-range dependencies** of input data, resulting in better coding processes [48].

G. CSI Feedback

With its powerful capabilities in learning data distribution and generating synthetic data, GAI has also been applied to recover compressed CSI feedback, as summarized in Table X. For example, the authors in [27] propose to use GAN for reconstructing CSI feedback in massive MIMO

TABLE X
SUMMARY OF GAI APPROACHES FOR CSI FEEDBACK, RADIO MAP ESTIMATION, AND CHANNEL DELAY ESTIMATION

Issue	Ref.	Drawbacks of TAI	Proposed GAI Approach
CSI feed-back	[27]	Cannot achieve performance as good as GAN	Use GAN to recover original CSI from its compressed version. Achieve an outdoor NMSE of -15.88 dB while CsiNet [139] and CsiNet+ [140] can only obtain -8.75 dB and -12.4 dB
	[142]	Cannot achieve performance as good as GAN	Use GAN to enhance wireless channel data, resulting in better CSI compression processes. Achieve a 3dB performance improvement compared to existing data augmentation approaches
	[141]	Less effective under noisy feedback channels	Use VAE to compress CSI under noisy channel conditions
Radio map estimation	[143]	<i>Lack of training data</i>	The generator aims to generate image masks while the discriminator learns to distinguish the masks of the original dataset and those generated by the generator.
	[144]	Poor performance due to nonuniformly positioned measurements and access constraints	Use conditional GAN architecture to efficiently estimate radio maps based on observations from the environment
Channel delay estimation	[145]	<i>Lack of training data</i>	Use GAN to generate synthetic cross-correlation data and smooth it with a Savitzky-Golay filter

communications systems. In particular, massive MIMO can provide high cell throughput and reduce multiuser interference but largely relies on exploiting the CSI feedback from UEs. To reduce the signaling overhead of the system, the CSI feedback is usually compressed at UEs before transmitting to BSs. During the compressing process, important CSI information may be removed unintentionally, resulting in low precoding performance at BSs. To tackle this problem, the authors develop a GAN-based CSI recovery framework that can effectively generate a CSI matrix based on its compressed version. Specifically, the compressed CSI feedback will be first fed to the generator to estimate the CSI vector. This estimated CSI vector is then fed to the discriminator together with the original CSI vector to determine if the reconstructed CSI is good or bad. A new loss function combining the adversarial loss of the discriminator and the mean square error loss between the reconstructed and original CSI is introduced to further enhance the recovery performance of the proposed GAN-based approach. Extensive simulations reveal that by using GAN, the proposed framework is superior to traditional DL-based approaches. For instance, with a compression ratio of $\frac{1}{4}$, the GAN-based framework can achieve an outdoor NMSE of -15.88 dB while CsiNet [139] and CsiNet+ [140] can only obtain -8.75 dB and -12.4 dB, respectively.

Differently, the authors in [141] propose to use VAEs for CSI compression at UEs under noisy channel conditions. The authors highlight that conventional DL-based CSI compression approaches such as CsiNet in [139] are vulnerable to noisy feedback channels which are common in practice. In contrast, the proposed VAE-based compressor can approximate distribution parameters for each dimension instead of estimating a point for each dimension in the latent space (i.e., deterministic latent space) as in classic DL-based solutions. As a result, the compressed CSI is robust against noise in the feedback channel. To make the proposed VAE network more suitable for the noise conditions of the feedback channel, the authors modify the VAE loss by using a weighted combination of reconstruction error and KL divergence between the encoder's distribution and the true distribution. The authors then test the proposed solution with an additive white Gaussian noise

(AWGN) feedback channel and indicate that the proposed VAE-based compression technique can outperform other DL-based techniques (e.g., CsiNet [139]) and compressive-sensing based models both under noise-free and noisy channel conditions. Similarly, the authors in [142] adopt GAN for wireless channel data augmentation before feeding CSI data into CsiNet. Specifically, a GAN-based network is developed to enrich data features of the original wireless channel data and also to generate new similar data by learning the distribution of the original channel data. These GAN-generated data will be fed into CsiNet for compression before feeding back to BSs. Simulation results reveal that using GAN can achieve a 3dB performance improvement compared to existing data augmentation approaches.

In summary, with the ability to learn data distribution and then **reconstruct data** from limited and noisy samples, GAI is a powerful tool for CSI feedback in wireless communications. Nevertheless, current applications are limited to GAN-based approaches. Further efforts in applying other advanced GAI techniques should be considered to reveal the full potential of GAI for CSI feedback.

H. Radio Map and Channel Delay Estimation

Due to its capability of variational learning and sampling to explore the data distribution in a more versatile manner, GAI can also be used for radio map estimation [143], [144], as summarized in Table X. In particular, a radio map spatially shows RF signal strength distribution and network coverage information which are essential characteristics for resource management and network planning in wireless communication systems. Unfortunately, conventional DL-based approaches such as RadioUNet [146] and autoencoder [147] may not be effective for radio map estimation in modern IoT and cellular systems due to nonuniformly positioned measurements and access constraints. For that, the authors in [143] and [144] propose to use the conditional GAN architecture to efficiently estimate radio maps based on observations from the environment. Particularly, the generator aims to generate image masks while the discriminator learns to distinguish the masks

of the original dataset and those generated by the generator. Simulation results then demonstrate the effectiveness of GAN in estimating radio maps in various outdoor environments.

In addition, GAN is a promising approach for channel delay estimation as studied in [145]. The authors aim to accurately estimate the first-arrival-path delay in wireless multi-path channels which plays an essential role in positioning and localization services. To do that, they first propose a CNN network to learn the mapping between the cross-correlation sequence and the delay offset. However, this CNN network suffers from the lack of training data. As such, the authors use GAN to generate synthetic cross-correlation data and smooth it with a Savitzky-Golay filter. The authors then perform various simulations to show that the proposed channel delay estimator can outperform existing approaches. In addition, the proposed GAN architecture can help to maintain a good estimation accuracy for the CNN network even with limited real cross-correlation data.

In conclusion, GAI has been mostly adopted for common issues in physical layer communications such as physical layer security, channel estimation, and signal detection. However, thanks to its capabilities, GAI can also be applied to other problems in the physical layer, such as radio map estimation, waveform generation, and channel delay estimation, opening new research directions.

IV. OPEN ISSUES AND FUTURE RESEARCH DIRECTIONS

Although having great capabilities in complex data feature extraction, transformation, and enhancement, GAI is still in its early stage of development. Thus, open issues and research directions of GAI in physical layer communications will be discussed in this section.

A. Security and Privacy

As discussed above, adversarial attacks can significantly impact GAI systems. In particular, adversaries can inject crafted perturbations into the input data of GAI models to replicate these models or degrade their performance. Moreover, GAI can be exploited by adversaries to generate data that is similar to legitimate/trusted data, making conventional security approaches less effective in classifying these adversarial attacks. However, there is limited effort in dealing with adversarial attacks, especially GAI-based attacks in physical layer communications. One potential approach is to *fight fire with fire* by using GAI models to generate adversarial training data and learn on this synthetic data to determine statistical anomalies that suggest potential perturbations. Moreover, GAI can be used to recover poisoned input data to mitigate the negative effects of adversarial perturbations [148].

B. Model-Driven GAI

As can be observed in Section III, existing GAI-based models mostly focus on data-driven approaches that rely on the availability of training data. However, in practice, collecting a sufficient amount of training data may be costly, time-consuming, and even impossible. To tackle this issue,

model-driven approaches [149], [150] can be adopted. In particular, model-driven approaches can incorporate the prior knowledge of target domains, e.g., carrier frequencies, physical constraints, and noise distributions, into the training process to further improve the performance of GAI-based solutions. For example, with prior knowledge of bandwidth and carrier frequency, GAI-based solutions can be trained to generate more realistic channel samples.

C. Resource-Efficient Learning

The training and inference of GAI require computation, storage, and communication resources, putting burdens on existing communication systems, especially for resource-constrained devices such as IoT devices, mobile phones, and UAVs. As such, novel GAI architectures need to be developed to minimize resource consumption while maintaining good learning performance. Distributed and federated learning can be integrated into GAI to offload computational tasks to edge devices as well as reduce communication overhead by transmitting model updates instead of raw data. For example, GAI models can be trained at edge devices with local data and then aggregated at a centralized server to obtain a global GAI model. In addition, GAI can be used to recover compressed local model updates to reduce communication overhead while still maintaining good training performance. Incentivization mechanisms such as dynamic spectrum access should also be considered to utilize communication resources, especially in cognitive radio networks as studied in a few papers reviewed in Section III.

D. Real-Time Adaptation

Although GAI has the capability of domain adaptation that can leverage knowledge from a source domain for training in a target domain, it still requires a large amount of training data and a long training time to achieve good performance. For example, when UAVs fly to a new area without prior channel knowledge, GAI models may not be able to quickly adapt to the new conditions with a limited number of labeled training samples. Consequently, GAI may not effectively deal with real-time wireless channel/environment changes caused by random factors such as mobility, blockage, and interference. For that, it is essential to develop novel GAI approaches that can quickly adapt to track these variations. Integrating advanced ML techniques such as meta-learning [151] into GAI is a promising direction to help GAI models quickly adapt to new environmental conditions based on a few training samples. Specifically, meta-learning can obtain important and useful information in the training process of source environments and use that knowledge to quickly learn new environments. With meta-learning, GAI can achieve high accuracy with a few training data samples in new wireless systems, making it more practical in real-world applications. In addition, over-the-air evaluation and implicit CSI feedback mechanisms should be developed to further improve the performance of GAI under the dynamics and uncertainty of physical layer communications.

V. CONCLUSION

Generative AI is a promising technology for physical layer communications due to its capabilities of complex data feature extraction, transformation, and enhancement. In this article, we have presented a comprehensive survey of the applications of generative AI in physical layer communications. Firstly, we have introduced an overview of generative AI, common generative models, and their advantages compared to traditional AI techniques. Then, we have provided detailed reviews, analyses, and comparisons of different generative AI techniques in emerging problems in physical layer communications such as channel modeling, channel estimation and signal detection, physical layer security, joint source channel coding, beamforming, and intelligent reflecting surface. Finally, we have highlighted important open issues and future research directions of generative AI in physical layer communications.

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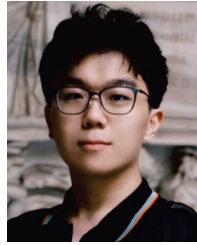
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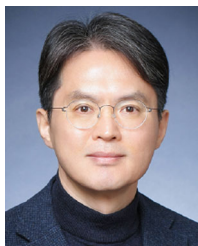


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